BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 042 · Computer Misuse Act 1990

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the purpose of the Data Protection Act 2018?

Q2. How many bytes are in 72 bits?
A. 18 bytes
B. 9 bytes
C. 72 bytes
D. 576 bytes

Q3. A text file contains 180 characters. The stated encoding uses 8 bits per character. Calculate its raw size in bits and bytes.

Q4. A microphone records digital sound. “Sampling stores every instant of the sound wave.” Correct this. How does it affect recorded detail?

I Do · Read new knowledge

Key vocabulary

Computer Misuse Act 1990 — UK law targeting unauthorised computer access and related misuse.

Unauthorised access — Entering or using computer material without permission.

Authorisation — Permission to access or use a system.

Computer offence — An action prohibited by criminal law involving computer systems.

Impairment — Damage or disruption to computer operation or data.

Penetration testing — Permission-based investigation of system vulnerabilities.

New learning

Purpose of the law: The Computer Misuse Act 1990 makes certain actions against computer systems unlawful, including access without permission.

Unauthorised access: Accessing computer material without authorisation can be an offence, even if the person does not steal information. Knowing a password does not by itself grant permission.

Further criminal intent: Unauthorised access with intent to commit or facilitate another offence is a more serious form of computer misuse.

Damaging or disrupting systems: Unauthorised acts that impair a system—such as deliberately causing harmful disruption or altering data without permission—can be offences. Authorised security testing must stay within a lawful agreed scope.

Worked example

A student guesses a teacher's password and opens restricted files without permission, but changes nothing. Explain why this may be an offence under the Computer Misuse Act 1990. [4 marks]

1. Identify the action: deliberately accessing another person's account.

2. The student gains access to computer material [1] without permission [1].

3. The Computer Misuse Act covers unauthorised access [1] even if no data is modified or deleted [1].

4. Do not incorrectly claim that a file must be damaged before unauthorised access becomes unlawful.

We Do · Practise with guidance

deleting or modifying data [1].

Q3. Unauthorised access is entering computer material without permission [2]. Access with intent to commit another offence adds criminal purpose [2]. Deliberate unauthorised acts impairing systems/data can disrupt operation [2]. Apply examples accurately without assuming legitimate security testing is unlawful.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It sets rules for using personal data lawfully and protects people's rights.

Q2. B — 9 bytes

Q3. 1,440 bits and 180 bytes.

Q4. Measure amplitude at regular intervals. Store each rounded (quantised) value in binary. Sample rate and bit depth limit the detail captured.

I Do / We Do pages · Guided practice

Q1. The employee uses an account to access restricted computer records [1] without being authorised [1]. Knowing a password is not permission [1], so the access may breach the Computer Misuse Act [1].

Q2. The changes were unauthorised [1] and impair the operation of the system [1]. This can amount to a Computer Misuse Act offence [1], and legitimate users may be unable to access the service/data [1].

Q3. Permission shows the tester is authorised [1] and distinguishes the agreed test from unauthorised access [1]. A defined scope identifies permitted systems and actions [1], preventing the tester from exceeding authorisation [1].

You Do pages · Code First

```
authorised = false
if authorised then
    print("Allow")
else
    print("Deny")
endif
```

Expected result: Deny

You Do pages · Independent questions

Q1. Accessing a computer/system without authorisation [1].

Q2. The account contains computer material [1], which the pupil deliberately accessed [1] without permission [1]. Unauthorised access is prohibited even without

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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Guided 1

An employee opens confidential HR records using a colleague's password without permission. Explain why knowing the password does not make the access lawful. [4 marks]

Worked method: The employee uses an account to access restricted computer records [1] without being authorised [1]. Knowing a password is not permission [1], so the access may breach the Computer Misuse Act [1].

Guided 2

A person deliberately changes a shared database so the service stops working. Explain which kind of computer misuse is involved and one effect. [4 marks]

Worked method: The changes were unauthorised [1] and impair the operation of the system [1]. This can amount to a Computer Misuse Act offence [1], and legitimate users may be unable to access the service/data [1].

Guided 3

A school hires an expert to test its website security. Explain TWO reasons written permission and a defined scope are important. [4 marks]

Worked method: Permission shows the tester is authorised [1] and distinguishes the agreed test from unauthorised access [1]. A defined scope identifies permitted systems and actions [1], preventing the tester from exceeding authorisation [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the simulated authorisation decision.

```
authorised = false
if authorised then
    print("Allow")
else
    print("Deny")
endif
```

predict

Question 1

State one action prohibited by the Computer Misuse Act 1990. [1 mark] Source: Original OCR J277-style teaching question; not an official OCR past paper.

Question 2

A pupil opens another pupil's school account without permission but does not delete any work. Explain why this can still be computer misuse. [4 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

Question 3

Compare unauthorised access, access with further criminal intent, and unauthorised damage/disruption under the Computer Misuse Act. [6 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

Accessing someone else's account without permission can be unlawful even if no files are deleted.